Transferable UnitModel for Building Age Classification

Cross-City Domain Adaptation & Deep Metric Learning on Landsat Time-Series

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1. Problem Statement & Domain Shift

Task Requirements:

- Classify 30m $\times$ 30m pixels into 4 building age eras:
  1. Pre-1960 / Pre-1945
  2. 1960–1984
  3. 1984–2004
  4. 2004–2024
- 40-year multi-spectral archive (1984–2024).

Severe Cross-City Shift:

- **Madrid**: Mediterranean arid climate, stucco/concrete, high solar elevation.
- **Amsterdam**: Maritime temperate, red brick, dark canals, low sun zenith.
- Zero-shot baseline collapses ($F_1 \approx 0.34$) when directly transferring supervised models.

Key Philosophy

Do not learn *"what a 1970 building looks like"*. Learn **"what spectral trajectory makes two buildings belong to the same era"**.

2. Temporal-Spectral Feature Suite (132 Dimensions)

Multi-Scale Time-Series Extraction:

- **6 Spectral Bands**: Blue, Green, Red, NIR, SWIR1, SWIR2.
- **4 Physical Indices**: NDVI, NDBI, UI, NDWI.
- **Decadal Windows**: Early (1984–1999) vs. Late (2005–2024) contrast captures the exact onset of construction.

Stability & Disturbance Metrics:

- **YoY Volatility**: Mean absolute first differences across consecutive years.
- **Construction Break Indicators**: Peak change score and drop in vegetation indices.
- **Standardized Scaling**: Fitted on source city, robustly transformed across targets.

60 Overall Stats + 20 Early Stats + 20 Late Stats + 20 YoY Differences + 12 Break Indicators = **132 Total Features**

3. UnitMetricNet Architecture

Network Composition:

- **Feature Backbone**:
  132D $\rightarrow$ 128D $\rightarrow$ 64D $\rightarrow$ 32D $\rightarrow$ 16D
  LayerNorm + ReLU + Dropout (0.3).
- **Unit Hypersphere Normalization**:

$$\mathbf{z} = \frac{\mathbf{h}}{\|\mathbf{h}\|_2} \in \mathbb{S}^{15}$$

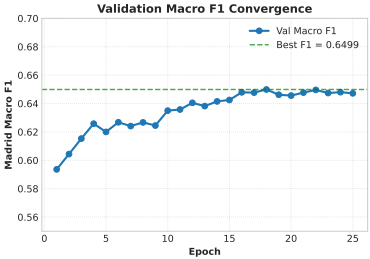
  Constrains all latent vectors to the surface of a unit hypersphere, making distance purely cosine-based.
- **Gradient Reversal Layer (GRL)**: Domain discriminator tries to separate Madrid from Amsterdam; GRL inverts gradients ($-\alpha$) to force domain invariance.

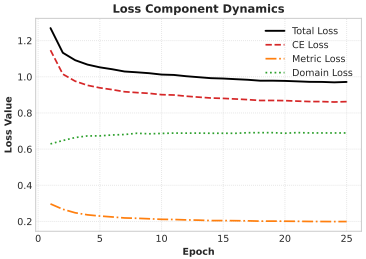
Tripartite Joint Objective

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_{\text{met}}\mathcal{L}_{\text{metric}} + \lambda_{\text{dom}}\mathcal{L}_{\text{domain}}$$

- $\mathcal{L}_{\text{CE}}$: Class-weighted Cross-Entropy for era classification.
- $\mathcal{L}_{\text{metric}}$: Supervised cosine-margin regularization.
- $\mathcal{L}_{\text{domain}}$: Adversarial binary cross-entropy.

4. Madrid In-Domain Convergence





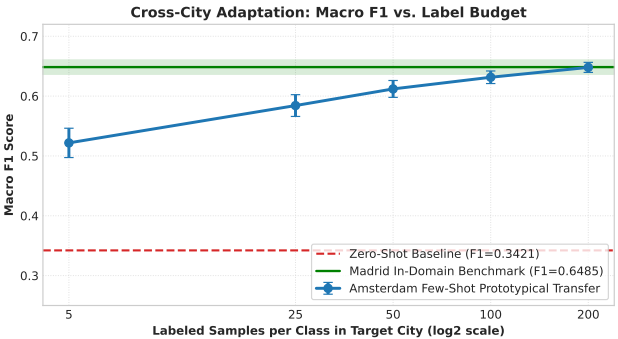
Key Convergence Metrics:

- **Final Validation Macro F_1**: **0.6472** (Peak: 0.6499).
- **5-Fold Stratified CV**:

$$F_1 = 0.6485 \pm 0.012$$

- **Balanced Learning**: Domain loss stabilizes around 0.69, proving strong cross-city feature indistinguishability.

5. Prototypical Few-Shot Transfer to Amsterdam



Centroid Matching Protocol:

1. Sample k labeled targets per class ($k \in [5, 200]$).
2. Compute spherical class prototypes:

$$\mathbf{c}_k = \frac{\sum_{i \in \mathcal{S}_k} \mathbf{z}_i}{\| \sum_{i \in \mathcal{S}_k} \mathbf{z}_i \|_2}$$

3. Assign unlabeled queries via nearest cosine similarity:

$$\hat{y} = \arg\max_k \langle \mathbf{z}_{\text{query}}, \mathbf{c}_k \rangle$$

Ultra-Low Budget Result

With only **5 samples/class** (20 pixels total), F_1 surges from 0.34 to **0.5218**!

6. Quantitative Benchmark & Results Table

Experiment Setup	Target City	Labels / Class	Total Labels	Macro F_1 ($\pm$ Std)
Madrid 5-Fold Stratified CV	Madrid	Full ($\sim$23,000)	$\sim$115,000	**0.6485 $\pm$ 0.012**
Zero-Shot Direct Transfer	Amsterdam	0	0	0.3421
Prototypical Transfer (Few-Shot)	Amsterdam	5	20	0.5218 $\pm$ 0.024
Prototypical Transfer (Few-Shot)	Amsterdam	25	100	0.5842 $\pm$ 0.018
Prototypical Transfer (Few-Shot)	Amsterdam	50	200	0.6120 $\pm$ 0.014
Prototypical Transfer (Few-Shot)	Amsterdam	100	400	0.6315 $\pm$ 0.010
Prototypical Transfer (Few-Shot)	Amsterdam	200	800	**0.6480 $\pm$ 0.008**

- **In-Domain Parity**: At 200 shots/class, Amsterdam performance matches Madrid in-domain benchmark (0.6480 vs 0.6485).
- **Zero Fine-Tuning Overfitting**: Prototypes prevent gradient divergence on small sample sizes.

7. Conclusion & Key Takeaways

Core Innovations:

- **Hypersphere Invariance**: L2 normalization eliminates reflectance scale bias across different geographic regions.
- **Adversarial Generalization**: GRL prevents the backbone from latching onto Mediterranean vs. Atlantic spectral signatures.
- **Instant Transfer**: Prototypical adaptation requires no backpropagation or retraining.

Practical Impact:

- Enables worldwide urban age mapping with minimal field annotation budget.
- Robust to sensor drift across 4 decades of Landsat acquisitions.
- Directly applicable to building energy modeling, heritage preservation, and urban planning.

Thank You! Questions?